

• Line 4:-

$$IN = \{a-d\}$$

$$OUT = \{a-d, b+c\}$$

$d = b+c$, $b+c$ already available from line 1 $\rightarrow$
Redundant computation

Optimized code:-

$$t = b+c$$

$$a = t$$

$$d = t$$

$$b = a-d$$

$$e = b+c$$

- $b+c$ is available at line 2 $\rightarrow$ reuse possible.
- After line 3, b changes $\rightarrow$ expression invalid.
- So, line 4 must recompute $b+c$.